Control System: Phase 2

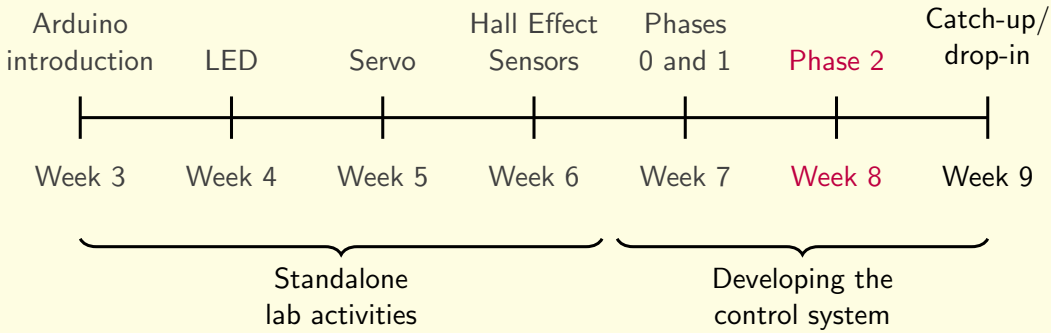
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MAC 233: **18/11/2026**



MAC 233: LABS OVERVIEW



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TODAY

Everything is built and tested. Today: how good is the example controller, and how could you make it better?

Not finished step 5 yet? Do that first.

Next week is a catch-up / drop-in session: come along if you need more time or help.

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TESTING THE EXAMPLE CONTROLLER

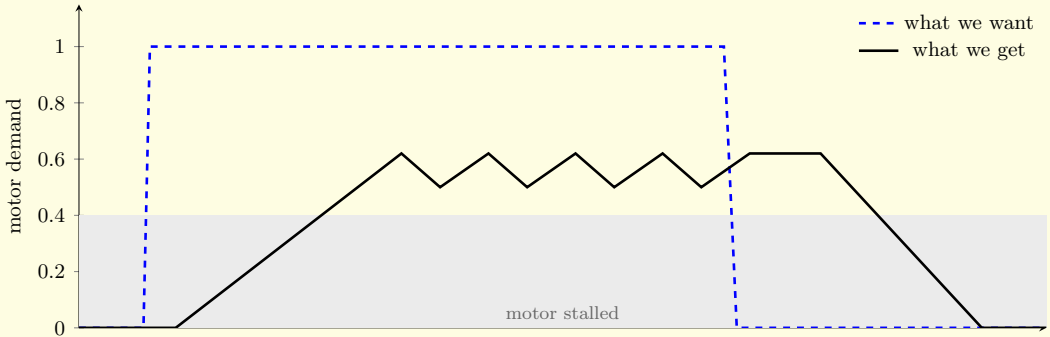
Predict: sketch the motor demand and the wheel speed against time.

Test: turn the crank about once a second for 20 seconds, then stop.

Compare: copy the serial monitor output and plot it. Does it match your sketch?

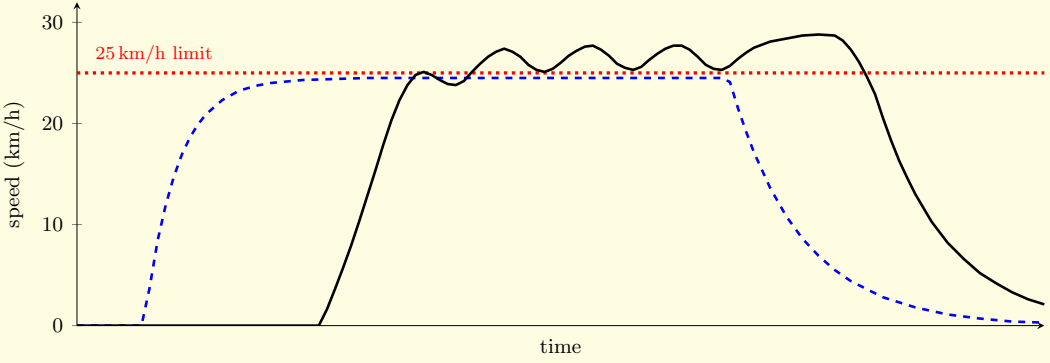
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WHAT WE WANT VS WHAT WE GET: MOTOR DEMAND



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WHAT WE WANT VS WHAT WE GET: SPEED



Dashed: what we want. Solid: what we get.

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THREE PROBLEMS

Slow start: the motor stalls until the demand reaches about 0.4.

Overshoot: the speed wobbles around the 25 km/h limit.

Late stop: the help carries on for up to 3 seconds after the rider stops.

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YOUR TURN

Change anything between the two lines of slashes in ‘loop’. Leave the `DO NOT CHANGE` block alone.

- Start the ramp from where your motor starts to turn.
- Ease off as the bike gets close to 25 km/h.
- Change the ramp rates.

Same method as Phase 1: one change at a time, predict, test, compare.

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HELP

Worksheet

Sketch

Problems? Raise your hand and someone will come and help you.

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